

Human Development Index (HDI) Predictor

Project Executable Files

Project: Human Development Index (HDI) Predictor

Phase: Project Documentation

1. Prerequisites

Install the required Python libraries:

```
pip install numpy pandas matplotlib seaborn scikit-learn flask
```

2. Step 1 — Regenerate the Dataset (optional)

```
cd Training
python generate_dataset.py
```

3. Step 2 — Train the Model

```
python train_model.py
# writes Flask/HDI.pkl and Flask/country_encoder.pkl
```

4. Step 3 — Run the Web Application

```
cd ../Flask
python app.py
# then open http://127.0.0.1:5000 in a browser
```

5. Executable Files Summary

File	Purpose
Training/generate_dataset.py	Builds Dataset/HDI.csv
Training/train_model.py	Trains and saves the model + encoder
Flask/app.py	Runs the web application